

WORKBOOK ASSIGNMENT-4

QUESTION1

Questions: Q1) Create a MySQL database named CollegeDB and create the following tables:

-- Create Database

```
CREATE DATABASE SeniorCollegeDB;
```

```
USE SeniorCollegeDB;
```

-- Create Student Table

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    City VARCHAR(30)  
);
```

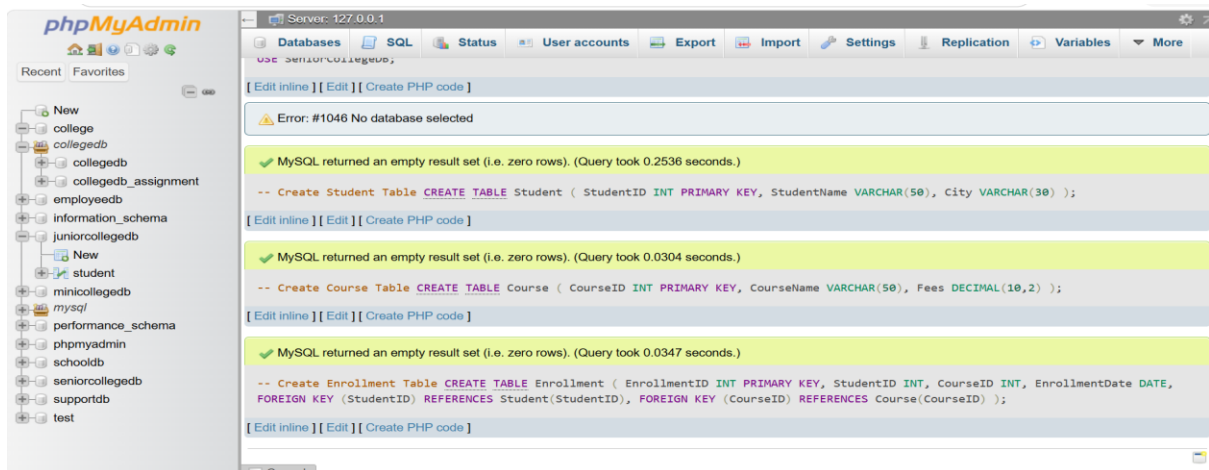
-- Create Course Table

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50),  
    Fees DECIMAL(10,2)  
);
```

-- Create Enrollment Table

```
CREATE TABLE Enrollment (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT,  
    CourseID INT,
```

EnrollmentDate DATE,
 FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
 FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);



QUESTION2

Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table. Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

-- Insert into Student table

INSERT INTO Student VALUES

(1, 'Alice', 'NY'),
 (2, 'Bob', 'LA'),
 (3, 'Charlie', 'SF'),
 (4, 'Dave', 'CHI'),
 (5, 'Eve', 'MIA'),
 (6, 'Frank', 'SEA');

-- Insert into Course table

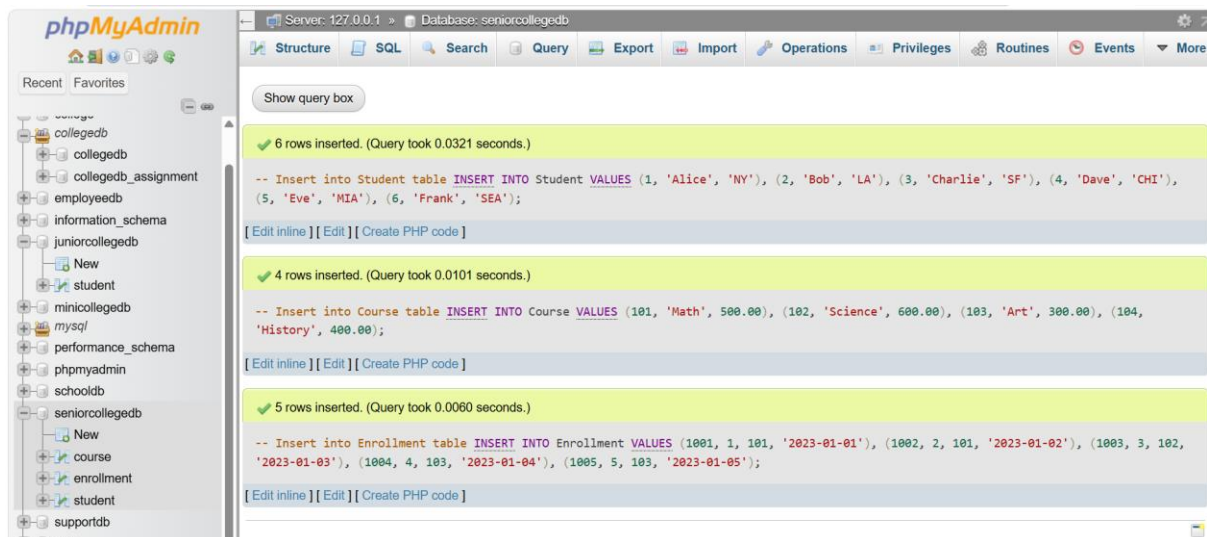
INSERT INTO Course VALUES

```
(101, 'Math', 500.00),
(102, 'Science', 600.00),
(103, 'Art', 300.00),
(104, 'History', 400.00);
```

-- Insert into Enrollment table

INSERT INTO Enrollment VALUES

```
(1001, 1, 101, '2023-01-01'),
(1002, 2, 101, '2023-01-02'),
(1003, 3, 102, '2023-01-03'),
(1004, 4, 103, '2023-01-04'),
(1005, 5, 103, '2023-01-05');
```



QUESTION3

Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses. The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

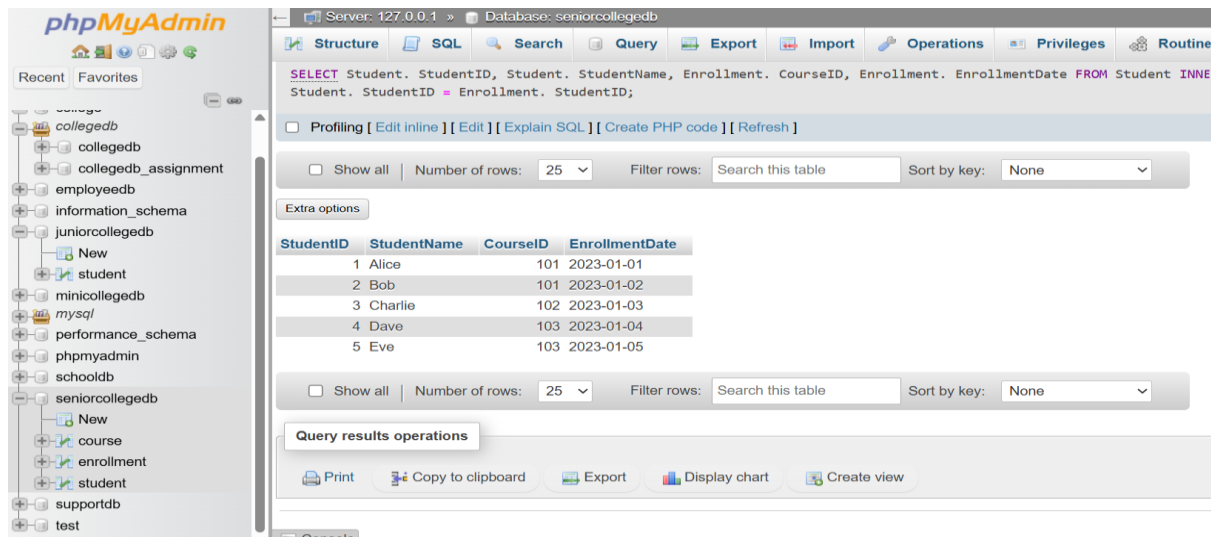
SELECT Student. StudentID,

Student. StudentName,

```

Enrollment. CourseID,
Enrollment. EnrollmentDate
FROM Student
INNER JOIN Enrollment ON
Student. StudentID =
Enrollment. StudentID;

```



QUESTION4

Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables. Use:

Tables Required
Student
Course
Enrollment

```

SELECT Student. StudentName,
student. city,
Course. CourseName,
Course. Fees,
Enrollment. EnrollmentDate
FROM Student
JOIN Enrollment ON
Student. StudentID =

```

Enrollment. StudentID
 JOIN Course ON Enrollment. CourseID
 = Course. CourseID;

The screenshot shows the phpMyAdmin interface for a database named 'seniorcollegedb'. The SQL query entered is:

```
SELECT Student. StudentName, student. city, Course. CourseName, Course. Fees, Enrollment. Enrollm
ON Student. StudentID = Enrollment. StudentID JOIN Course ON Enrollment. CourseID = Course. Cours
```

The query results are displayed in a table with the following data:

StudentName	city	CourseName	Fees	EnrollmentDate
Alice	NY	Math	500.00	2023-01-01
Bob	LA	Math	500.00	2023-01-02
Charlie	SF	Science	600.00	2023-01-03
Dave	CHI	Art	300.00	2023-01-04
Eve	MIA	Art	300.00	2023-01-05

QUESTIONS

Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course. The output should include:

Column
StudentID
StudentName
City
CourseID
EnrollmentDate

SELECT Student. StudentID,
 Student. StudentName,
 Student. City,
 Enrollment. CourseID,
 Enrollment. EnrollmentDate
 FROM
 Student
 LEFT JOIN Enrollment
 ON Student.StudentID = Enrollment.StudentID;

Server: 127.0.0.1 » Database: seniorcollegedb

Enrollment ON Student.StudentID = Enrollment.StudentID;

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID	StudentName	City	CourseID	EnrollmentDate
1	Alice	NY	101	2023-01-01
2	Bob	LA	101	2023-01-02
3	Charlie	SF	102	2023-01-03
4	Dave	CHI	103	2023-01-04
5	Eve	MIA	103	2023-01-05
6	Frank	SEA	NULL	NULL

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Console

QUESTION6

Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students. The output should include:

Column
StudentName
CourseID
CourseName
Fees

```

SELECT Student.StudentName,
Course. CourseID,
Course. CourseName,
Course. Fees
FROM Student
JOIN Enrollment ON
Student.StudentID = Enrollment.StudentID
RIGHT JOIN Course ON
Enrollment. CourseID = Course. CourseID;

```

phpMyAdmin

Recent

Favorites

collegedb

collegedb

collegedb_assignment

employeedb

information_schema

juniorcollegedb

New

student

minicollegedb

mysql

performance_schema

phpmyadmin

schooldb

seniorcollegedb

New

course

enrollment

student

supportdb

test

Server: 127.0.0.1 » Database: seniorcollegedb

Structure

SQL

Search

Query

Export

Import

Operations

Privileges

Routing

Enrollment.StudentID RIGHT JOIN Course ON Enrollment.CourseID = Course.CourseID;

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

StudentName	CourseID	CourseName	Fees
Alice	101	Math	500.00
Bob	101	Math	500.00
Charlie	102	Science	600.00
Dave	103	Art	300.00
Eve	103	Art	300.00
NULL	104	History	400.00

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view